IT, Comp, DS, AI

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Academic Year: 2023-2024

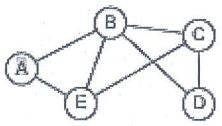
Program/s: **B Tech/ MBA Tech**	Year: 1 Semester:II
Stream/s ;B Tech IT, CE, CS, EXTC, MXTC, DS, CSE-CS, AI, AIML, AIDS/MBA Tech IT, COMP, DS, AI	
Subject: Data Structures and Algorithms	Time: 3 hrs (2.00pm to 5.00 pm)
Date: 22 / 4 / 2024	No. of Pages: 2
Marks: 100	

Re-Examination (2021-22)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. 1 is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5 questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	[20]
CO- ; SO- ; BL-	a.	Explain performance parameters: time and space complexity Calculate the time complexity for the given algorithm: Algorithm sum(n) 1. int s=0 2. for i=1 to n 3. s=s+I; 4. Print s	
CO- ; SO- ; BL-	b.	Differentiate arrays and linked list	
CO- ; SO- ; BL-	c.	Construct a binary search tree for the numbers 9, 2, 1, 8, 3, 6, 0, 7, 4, 5. Assume that the numbers are inserted in that order into an initially empty binary search tree.	
CO- ; SO- ; BL-	d.	Sort the elements 77, 49, 25, 12, 9 using selection sort. Show all the passes.	
Q2 CO-; SO-;	a.	Write an algorithm to perform insert and delete operations on a circular queue data structure?	[10]

BL-			
	b	Write an algorithm to perform following operations on a singly linked list. a. Insert a new node at the end of the linked list b. Delete the last node.	[10]
Q3 CO-; SO-; BL-	a.	Construct the AVL Tree for the given set of numbers 21, 26, 30, 9, 4, 14, 28, 18,15,10, 2, 3, 7.	[10]
	b.	Write an algorithm to sort a given set of numbers using merge sort.	[10]
Q4 CO-; SO-; BL-	a.	Explain Graph data structure with example. Perform Depth first search of the given graph using DFS traversal algorithm (start vertex is A). 	[10]
	b.	Explain Heap sort algorithms. Sort the given array={4,10,3,5,1} using Heap sort algorithm.	[10]
Q5 CO-; SO-; BL-	a.	Write an algorithm to convert infix to postfix expression?	[10]
	b.	Perform following search algorithms with example i. Linear Search ii. Binary Search	[10]
Q6 CO-; SO-; BL-	a.	Write algorithm to perform following operations on a binary tree: i. In-order traversal ii. Pre-order traversal iii. Post-order traversal	[10]
	b.	Consider a hash table with size = 10. Using linear probing, insert the keys 27, 72, 63, 42, 36, 18, 29, and 101 into the table.	[10]
Q7 CO-; SO-; BL-	a.	Write short notes on: i. B+ Trees ii. Threaded binary tree	[10]
	b.	Explain various types of file access methods.	[10]